

Solver Documentation
electroChem
carbonEq branch

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1 Introduction

This document documents the custom OpenFOAM solver electroChem made for simulating reacting multicomponent species electrolytic systems with the goal of supporting microbial electrosynthesis of CO₂ to acetate. The solver is an extension of the existing pimpleFOAM FVM solver from OpenFOAM.

2 carbonEq branch

This branch adds an equilibrium projection model for the combined water and inorganic carbon equilibrium. This model aims to resolve the equilibrium species distribution in a way that satisfies mass conservation and electro-neutrality.

3 Governings equations

The flow is treated as incompressible and solved through the incompressible Navier Stokes formulation using the PIMPLE algorithm

$$\nabla \cdot \mathbf{u} = 0 \quad (1)$$

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) = -\frac{1}{\rho} \nabla P + \nu \nabla^2 \mathbf{u} + \mathbf{g} \quad (2)$$

All active species are treated by reactive Nerst-Planck

$$\frac{\partial c_i^*}{\partial t} + \nabla \cdot (c_i^* \mathbf{u}) = \nabla \cdot (D_i \nabla c_i^*) + \nabla \cdot \left(D_i \frac{z_i F}{RT} c_i^* \nabla \phi \right) + R_i \quad (3)$$

Where c_i^* is the transported species concentration, R_i is the reactive species source term combining the contributions of all reactions

$$R_i = \sum_j \nu_{ij} \dot{r}_j \quad (4)$$

where ν_{ij} is the stoichiometric constant of species "i" for reaction "j". After transport fast equilibrium reactions are assumed to instantaneously settle giving projected equilibrium concentration c_i^{eq} from which the potential is update. If no equilibrium reactions are included, then $c_i^{eq} = c_i^*$. A constant current density is prescribed over the domain in the electrode normal direction $\mathbf{e}_J$

$$\mathbf{j} = j_{app} \cdot \mathbf{e}_J \quad (5)$$

With this the potential gradient can be algebraic updated from the current equation

$$\mathbf{j} = -F \sum_i z_i D_i \nabla c_i^{eq} - \frac{F^2}{RT} \sum_i z_i^2 D_i c_i^{eq} \nabla \phi \quad (6)$$

with

$$\kappa = \frac{F^2}{RT} \sum_i z_i^2 D_i c_i^{eq} \quad (7)$$

and

$$\mathbf{j}_D = -F \sum_i z_i D_i \nabla c_i^{eq} \quad (8)$$

resulting in

$$\nabla \phi = \frac{\mathbf{j}_D - \mathbf{j}}{\kappa} = \frac{\mathbf{j}_D - j_{app} \cdot \mathbf{e}_J}{\kappa} \quad (9)$$

Species are converted to mass fraction

$$Y_i = c_i^{eq} \frac{M_i}{\rho} \quad (10)$$

and the inert species "I" is explicitly found by

$$Y_I = 1 - \sum_{i \neq I} Y_i \quad (11)$$

pH is the log concentration of H^+ in mol/L

$$pH = -\log_{10} \left(\frac{c_{H^+}}{1000} \right) \quad (12)$$

4 Custom boundary conditions

The electrode flux is comprised of a faradaic source combined with diffusive and migration flux assuming no penetration of the electrode.

$$\nabla c_i \cdot \mathbf{n} = \frac{\nu_i j_{el}}{D_i n F} - \left(\frac{z_i F}{RT} c_i \nabla \phi \right) \quad (13)$$

Discretization

The boundary is solved using a Robin equation formulation. Introduce

$$A = \frac{\nu_i j}{n F D_i}, \quad \beta = \frac{z_i F}{RT} (\nabla \phi \cdot \mathbf{n}) \quad (14)$$

At the electrode boundary

$$\nabla c_i \cdot \mathbf{n} = A - \beta c_{b,i} \quad (15)$$

where $c_{b,i}$ is the boundary concentration of species i . Then the boundary adjacent gradient is defined as

$$\nabla c_i = \delta (c_{b,i} - c_{p,i}) \quad (16)$$

where $\delta = 1/\Delta_n$, Δ_n is the distance from boundary to cell-center and $c_{p,i}$ is the boundary adjacent cell-center concentration. Insert into equation 15

$$\delta (c_{b,i} - c_{p,i}) = A - \beta c_{p,i} \quad (17)$$

isolate boundary concentration

$$c_{b,i} = \frac{\delta c_{p,i} + A}{\delta + \beta} \quad (18)$$

then the boundary flux becomes

$$\nabla c_i \cdot \mathbf{n} = A - \beta c_{b,i} \quad (19)$$

5 Reaction Models

The solver supports kinetic reactions treated as source terms or instantaneous equilibrium projections applied after species transport. When combining slow species consumption with fast equilibrium reactions, consumption can cause small negative concentration overshoots by temporarily exceeding the locally available concentration. When the equilibrium is much faster than the consumption process, it should act as a source that supplies this deficit. This can occur, for example, when electrode boundaries explicitly consume a reactant. Negative overshoots generated in this way should therefore be retained by the fast equilibrium model rather than clipped, which is why some equilibrium models allow small negative overshoots.

5.1 Kinetic Reactions

Kinetic reactions are for reactions developing on timescales comparable or slower than that of the transport and must be treated explicitly.

5.1.1 Reversible kinetic reaction

The reaction rate of reversible kinetic reactions are found as

$$\dot{r} = k_f \prod c_R - k_b \prod c_P \quad (20)$$

where k_f is the forward reaction rate and k_b is the reverse reaction rate. c_R is the concentration of reactants and c_P is the concentration of products. This model can allow for negative concentrations when used for fast equilibrium models.

5.1.2 Multiplicative Monod reaction

Microbial reactions are described through multiplicative Monod

$$\dot{r} = X \dot{q}_{max} \left(\frac{c_d}{c_d + K_{S.d}} \right) \left(\frac{c_a}{c_a + K_{S.a}} \right) \quad (21)$$

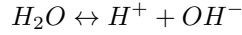
where X is the biomass density, $\dot{q}_{max}$ is the maximum specific consumption rate, K_S is the half saturation constant and the subscripts d and a describe the electron donor and acceptor substrates.

5.2 Equilibrium Reactions

Equilibrium reactions assume that the equilibrium settles much faster than the transport time scale and can therefore be assumed instantaneously before the next transport step.

5.2.1 Water equilibrium

The water equilibrium model projects the local water equilibrium under the constraint that the local ion difference is preserved as the reaction can only happen for ion pairs. The equilibrium reaction of water is



with equilibrium constant

$$K_W = c_{H^+} c_{OH^-} \quad (22)$$

Assuming water is treated as an inert species and does not actively participate. The ion difference is defined as

$$A = c_{H^+} - c_{OH^-} \quad (23)$$

Rewriting and substituting equation 23 into equation 28 the following algebraic concentration projections are found

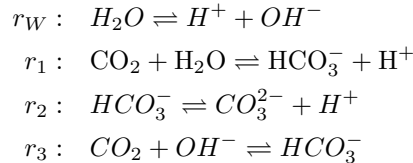
$$c_{H^+} = \frac{A + \sqrt{A^2 + 4K_w}}{2} \quad (24)$$

$$c_{OH^-} = \frac{-A + \sqrt{A^2 + 4K_w}}{2} \quad (25)$$

This gives the local water equilibrium but does not enforce any electro-neutrality conditions. This model can allow for negative concentrations for compensating for overshooting slow consumptions, such as electrode consumption.

5.2.2 Carbon water equilibrium

This model treats the combined water and carbonate species equilibria using carbon conservation and electro-neutrality as closure models. This treats the fast equilibrium reactions as instantaneously equilibrated after each transport step, effectively letting the reactions enforce electro-neutrality. This model assumed water be treated as inert. For an aqueous system saturated with CO_2 the following simultaneous equilibrium reaction governs the buffer



See A.1 for full derivation of the reaction system and treatment of slow kinetics. The slow reactions r_1 and r_3 are treated as explicit kinetic reactions

$$\dot{r}_1 = \frac{k_{f.1}}{K_{1.2}} c_{HCO_3^-} c_{H^+} - k_{b.1} c_{CO_2} \quad (26)$$

$$\dot{r}_3 = k_{f.3} c_{HCO_3^-} - k_{b.3} c_{CO_2} c_{OH^-} \quad (27)$$

and the fast equilibrium reactions r_W and r_2 are solved using their equilibrium equations

$$K_W = c_{H^+} c_{OH^-} \quad (28)$$

$$K_2 = \frac{c_{CO_3^{2-}} c_{H^+}}{c_{HCO_3^-}} \quad (29)$$

This results in 4 variables c_{H^+} , c_{OH^-} , $c_{HCO_3^-}$ and $c_{CO_3^{2-}}$ but only two equilibrium equations. Therefore, 2 closure models are needed. The first is the conservation of inorganic carbon

$$C_T = c_{HCO_3^-} + c_{CO_3^{2-}} \quad (30)$$

And the second is electro-neutrality, enforcing no net charge in the bulk

$$\sum_i z_i c_i = 0 = c_{H^+} - c_{OH^-} - c_{HCO_3^-} - 2c_{CO_3^{2-}} + c_{constant} \quad (31)$$

where $c_{constants}$ is the contribution from charged species not participating in the equilibrium system.

Numerical treatment

Rewriting equations (28-31) the equilibrium system can be reduced down to a single variable c_{H^+} . From 28 c_{OH^-} is rewritten as:

$$K_W = c_{H^+} c_{OH^-} \Rightarrow c_{OH^-} = \frac{K_W}{c_{H^+}} \quad (32)$$

From equation (29) $c_{HCO_3^-}$ and $c_{CO_3^{2-}}$ are rewritten as

$$K_2 = \frac{c_{CO_3^{2-}} c_{H^+}}{c_{HCO_3^-}} \Rightarrow \begin{cases} c_{HCO_3^-} &= \frac{c_{CO_3^{2-}} c_{H^+}}{K_2} \\ c_{CO_3^{2-}} &= \frac{K_2 c_{HCO_3^-}}{c_{H^+}} \end{cases} \quad (33a)$$

The resulting expressions from (33a-33b) are then inserted into equation 30. This gives a new expression for $c_{HCO_3^-}$

$$C_T = c_{HCO_3^-} + \frac{K_2 c_{HCO_3^-}}{c_{H^+}} \Rightarrow c_{HCO_3^-} = C_T \left(\frac{c_{H^+}}{c_{H^+} + K_2} \right) \quad (34)$$

and for $c_{CO_3^{2-}}$

$$C_T = \frac{c_{CO_3^{2-}} c_{H^+}}{K_2} + c_{CO_3^{2-}} \Rightarrow c_{CO_3^{2-}} = C_T \left(\frac{K_2}{K_2 + c_{H^+}} \right) \quad (35)$$

The new expressions for c_{OH^-} , $c_{HCO_3^-}$ and $c_{CO_3^{2-}}$ are inserted into equation (31) to form a single non-linear scalar equation for c_{H^+}

$$f(c_{H^+}) = c_{H^+} - \frac{K_W}{c_{H^+}} - C_T \left(\frac{c_{H^+} + 2K_2}{c_{H^+} + K_2} \right) + c_{constant} = 0 \quad (36)$$

From this c_{H^+} can be solved numerically using the Newton's method. This requires the derivative of (36):

$$f'(c_{H^+}) = \frac{\partial f}{\partial c_{H^+}} = 1 + \frac{K_W}{c_{H^+}^2} + C_T \frac{K_2}{(c_{H^+} + K_2)^2} \quad (37)$$

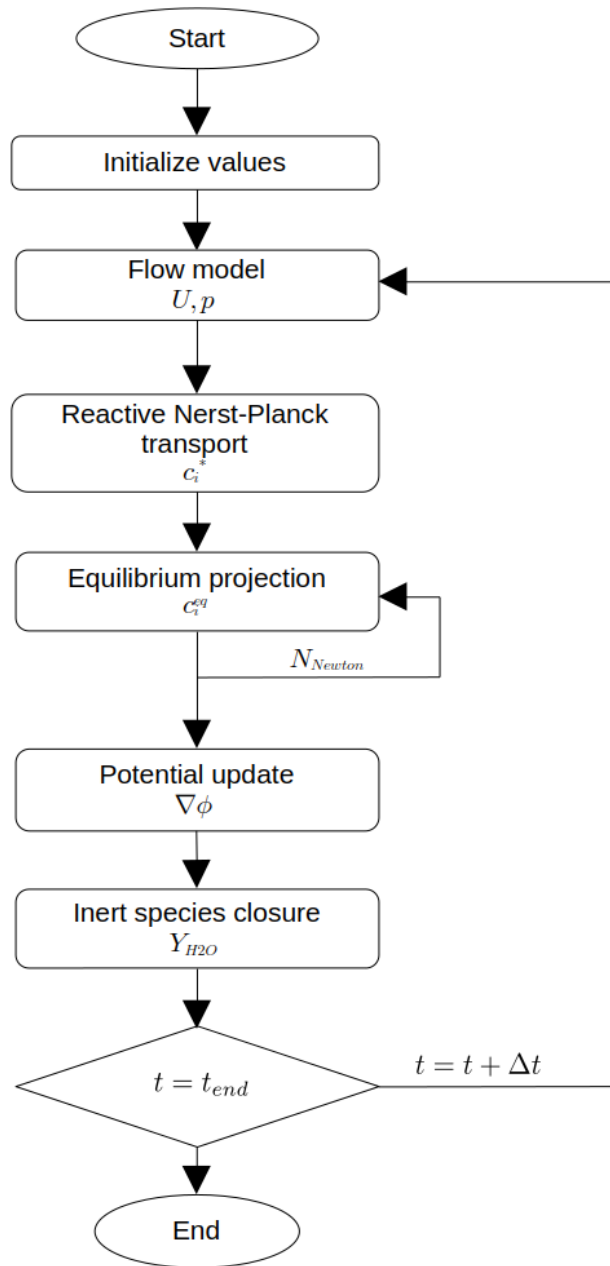
Then the concentration is updated using

$$c_{H^+}^{n+1} = c_{H^+}^n - \frac{f(c_{H^+}^n)}{f'(c_{H^+}^n)} \quad (38)$$

This is done for "N" corrector steps where "n" is the current corrector step. The transported quantity $c_{H^+}^*$ will serve as the initial guess. From the converged value $c_{H^+}^{eq}$ equations (32, 33a and 33b) are used to find the remaining equilibrium concentrations

$$\begin{aligned} c_{OH^-}^{eq} &= \frac{K_W}{c_{H^+}^{eq}} \\ c_{HCO_3^-}^{eq} &= C_T \left(\frac{c_{H^+}^{eq}}{c_{H^+}^{eq} + K_2} \right) \\ c_{CO_3^{2-}}^{eq} &= C_T \left(\frac{K_2}{K_2 + c_{H^+}^{eq}} \right) \end{aligned}$$

6 Algorithm flowchart



7 Solver structure

Source code structure

```
electroChem/  
|-- Documentation/  
|-- Make/  
|   |-- files  
|   |-- options  
|-- reactionModels/  
|   |-- equilibriumReactions/  
|       |-- waterEquilibrium.C  
|       |-- waterEquilibrium.H  
|       |-- waterCarbonEquilibrium.C  
|       |-- waterCarbonEquilibrium.H  
|   |-- kineticReactions/  
|       |-- multiplicativeMonod.C  
|       |-- multiplicativeMonod.H  
|       |-- reversibleKineticReaction.C  
|       |-- reversibleKineticReaction.H  
|-- cEqn.H  
|-- createFields.H  
|-- diagnostics.H  
|-- electroChem.C  
|-- pEqn.H  
|-- phiEqn.H  
|-- UEqn.H  
|-- YEqn.H
```

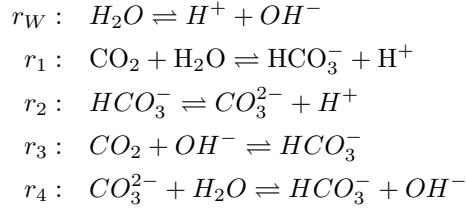
custom functions

```
src/  
|-- customLibs/  
|   |-- electrochemistry/  
|       |-- derivedFvPatchFields/  
|           |-- MolarFlux/  
|               |-- faradaicMolarFlux.C  
|               |-- faradaicMolarFlux.H  
|               |-- robinMolarFlux.C  
|               |-- robinMolarFlux.H  
|-- Make/  
|   |-- files  
|   |-- options
```

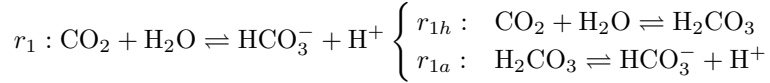
A Appendix

A.1 Carbon water equilibrium

Full reaction scheme



The CO_2 -bicarbonate equilibrium r_1 consists of a lumped hydration step $r_{1.1}$ and acid dissociation step $r_{1.2}$



The CO_2 hydration step as well as the hydroxide path r_3 are both slow reactions and cannot comfortably be treated as instantaneous equilibrium projections and must be treated kinetically

$$\dot{r}_{1h} = k_{f1}c_{H_2CO_3} - k_{b1}c_{CO_2} \quad (39)$$

$$\dot{r}_3 = k_{f.3}c_{HCO_3^-} - k_{b.3}c_{CO_2}c_{OH^-} \quad (40)$$

reactions $r_{1.h}$ and $r_{1.a}$ can be thermodynamically lumped into a single kinetic equation.

$$K_{1.2} = \frac{c_{HCO_3^-}c_{H^+}}{c_{H_2CO_3}} \Rightarrow c_{H_2CO_3} = \frac{c_{HCO_3^-}c_{H^+}}{K_{1.2}} \quad (41)$$

$$\dot{r}_1 = \frac{k_{f.1}}{K_{1.2}}c_{HCO_3^-}c_{H^+} - k_{b.1}c_{CO_2} \quad (42)$$

Effectively eliminating H_2CO_3 as a variable and making CO_2 treated entirely kinetically. Then the renaming equilibrium reactions are

$$K_W = c_{H^+}c_{OH^-} \quad (43)$$

$$K_2 = \frac{c_{CO_3^{2-}}c_{H^+}}{c_{HCO_3^-}} \quad (44)$$

$$K_4 = \frac{c_{HCO_3^-}c_{OH^-}}{c_{CO_3^{2-}}} \quad (45)$$

Where r_4 can be shown to be redundant. The equilibrium equations for water and bicarbonate-carbonate are rewritten

$$K_2 = \frac{c_{CO_3^{2-}}c_{H^+}}{c_{HCO_3^-}} \Rightarrow \frac{c_{HCO_3^-}}{c_{CO_3^{2-}}} = \frac{c_{H^+}}{K_2} \quad (46)$$

$$K_W = c_{H^+}c_{OH^-} \Rightarrow c_{OH^-} = \frac{K_W}{c_{H^+}} \quad (47)$$

The rewritten forms are multiplied and H^+ cancel out on the right-hand side

$$\frac{c_{HCO_3^-}c_{OH^-}}{c_{CO_3^{2-}}} = \frac{c_{H^+}K_W}{K_2c_{H^+}} = \frac{K_W}{K_2} \quad (48)$$

Showing the left-hand side is exactly the equilibrium constant for r_4

$$K_4 = \frac{K_W}{K_2} \quad (49)$$

making it a linear combination of the other equilibrium reactions and thereby redundant as they are not resolved kinetically but projected.